GOWIN
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Gowin DVI TX RX IP

User Guide

IPUG938-2.4E, 01/17/2025

Revision History

Date	Version	Description
04/15/2020	1.0E	Initial version published.
10/30/2020	2.0E	● TLVDS_OBUF and ELVDS_OBUF options added in DVI TX IP. ● Use External PLL option added in DVI RX IP.
02/26/2021	2.1E	Supported device added.
09/18/2021	2.2E	Phase Search Mode and Debug added in DVI RX IP.
10/31/2023	2.3E	Descriptions of Phase Search Mode added.
01/17/2025	2.4E	● O_pll_lock output port added. ● Auto mode code updated.

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1 About This Guide

1.1 Purpose

The purpose of Gowin DVI TX RX IP User Guide is to help you learn the features and usage of Gowin DVI TX RX IP by providing the descriptions of the product overview, functional description, GUI and reference design. The software screenshots in this manual are based on V1.9.11. As the software is subject to change without notice, some information may not remain relevant and may need to be adjusted according to the software that is in use.

1.2 Related Documents

You can find the related documents at www.gowinsemi.com:

- DS100, GW1N series of FPGA Products Data Sheet
- DS117, GW1NR series of FPGA Products Data Sheet
- DS821, GW1NS series of FPGA Products Data Sheet
- DS861, GW1NSR series of FPGA Products Data Sheet
- DS881, GW1NSER series of SecureFPGA Products Data Sheet
- DS891, GW1NRF series of Bluetooth FPGA Products Data Sheet
- DS102, GW2A series of FPGA Products Data Sheet
- DS226, GW2AR series of FPGA Products Data Sheet
- DS961, GW2ANR series of FPGA Products Data Sheet
- DS976, GW2AN-55 Data Sheet
- DS971, GW2AN-18X & 9X Data Sheet
- DS981, GW5AT series of FPGA Products Data Sheet
- DS1103, GW5A series of FPGA Products Data Sheet
- DS1239, GW5AST series of FPGA Products Data Sheet
- DS1105, GW5AS series of FPGA Products Data Sheet
- DS1108, GW5AR series of FPGA Products Data Sheet
- DS1118, GW5ART series of FPGA Products Data Sheet

- SUG100, Gowin Software User Guide

1.3 Terminology and Abbreviations

Table 1-1 shows the abbreviations and terminology used in this manual.

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Meaning
DDWG	Digital Display Working Group
DE	Data Enable
DVI	Digital Visual Interface
FPGA	Field Programmable Gate Array
HS	Horizontal Sync
IP	Intellectual Property
RGB	R(Red) G(Green) B(Blue)
SRAM	Static Random Access Memory
TMDS	Transition Minimized Differential Signaling
VESA	Video Electronics Standards Association
VS	Vertical Sync

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly by the following ways.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 Overview

2.1 Overview

DVI (Digital Visual Interface) transmits digital signals based on TMDS (Transition Minimized Differential Signaling). DVI TX IP is used to receive parallel video signals, which are then encoded into TMDS signals according to the DVI specification. DVI RX IP is used to receive TMDS signals and then decodes them into parallel video signals according to the DVI specification.

Table 2-1 Gowin DVI TX RX IP Overview

Gowin DVI TX RX IP	
Logic Resource	Refer to Table 2-2 and Table 2-3.
Delivered Doc.	
Design Files	Verilog (encrypted)
Reference Design	Verilog
TestBench	Verilog
Test and Design Flow	
Synthesis Software	GowinSynthesis
Application Software	Gowin Software (V1.9.5.02Beta and above)

Note!

For the devices supported, you can click here to get the information.

2.2 Features

- Supports DVI 1.0 specification
- Supports DVI-D interface.
- Supports Single-link TMDS
- Supports low-voltage differential signaling
- The data rate of single differential channel ranges from 80Mb/s~800Mb/s

2.3 Resource Utilization

DVI TX RX can be implemented by Verilog. Its performance and resource utilization may vary when the design is employed in a different device, or at a different density, speed, or grade. Take GW1N-4 as an instance, and the resource utilization is as shown in Table 2-2 and Table 2-3.

Table 2-2 DVI TX Resource Utilization

Series	Speed Grade	Name	Resource Utilization	Remarks
GW1N-4	-6	LUT	335	Configure internal PLL
		REG	91	
		PLL	1	
		OSER10	4	

Table 2-3 DVI RX Resource Utilization

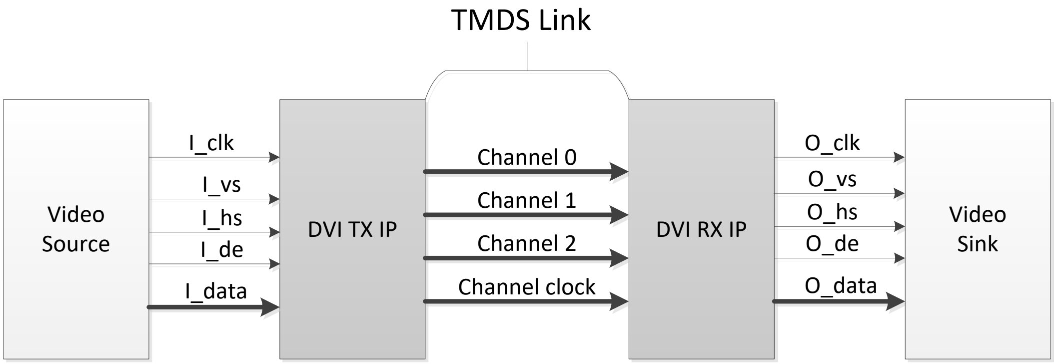
Series	Speed Grade	Name	Resource Utilization	Remarks
GW1N-4	-6	LUT	902	Configure internal PLL
		REG	427	
		PLL	1	
		CLKDIV	1	
		IODELAY	3	
		IDES10	3	

3 Functional Description

3.1 System Diagram

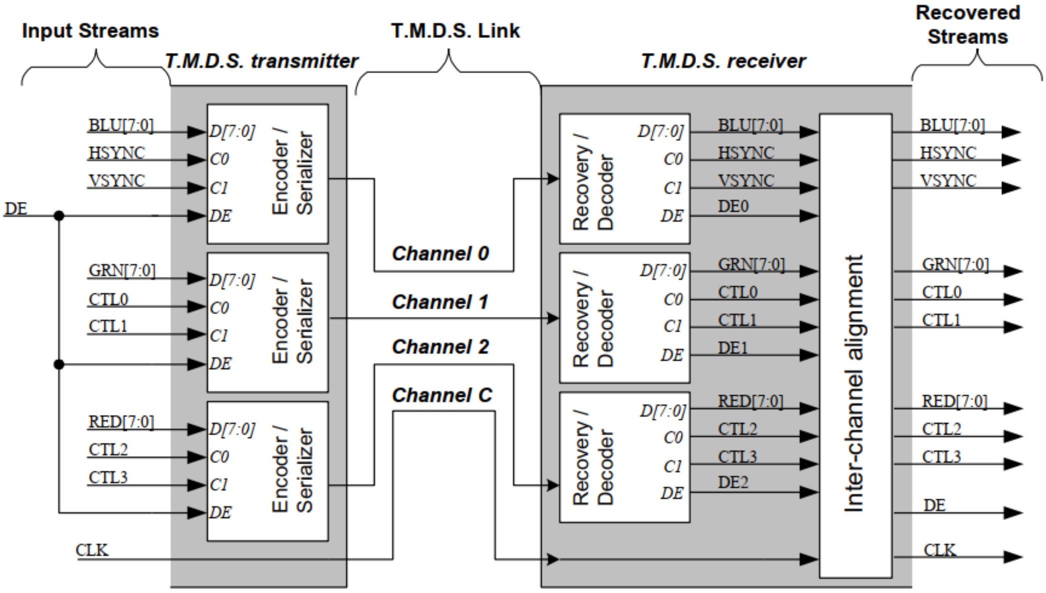
Gowin DVI IP includes DVI TX IP and DVI RX IP, and its system diagram is as shown in Figure 3-1.

Figure 3-1 System Block Diagram



3.2 Implementation Diagram

Figure 3-2 Implementation Diagram



Gowin DVI TX IP includes encoder and serializer modules. Gowin DVI RX IP includes decoder, recovery and inter-channel alignment modules.

3.2.1 DVI TX

According to the DVI specification, there are three modules in the single-link TMDS transmitter. The correspondence between each module and each component of the video signal is shown in Figure 3-2. Channel 0 corresponds to blue and includes HSYNC and VSYNC. Channel 1 corresponds to green and channel 2 corresponds to red. CTL0, CTL1, CTL2 and CTL3 must be set to 0.

The algorithm used by the modules is as shown in Figure 3-3, and the definitions of the signals in the diagram are as shown in Table 3-1.

Figure 3-3 TMDS Encoding Algorithm Flow

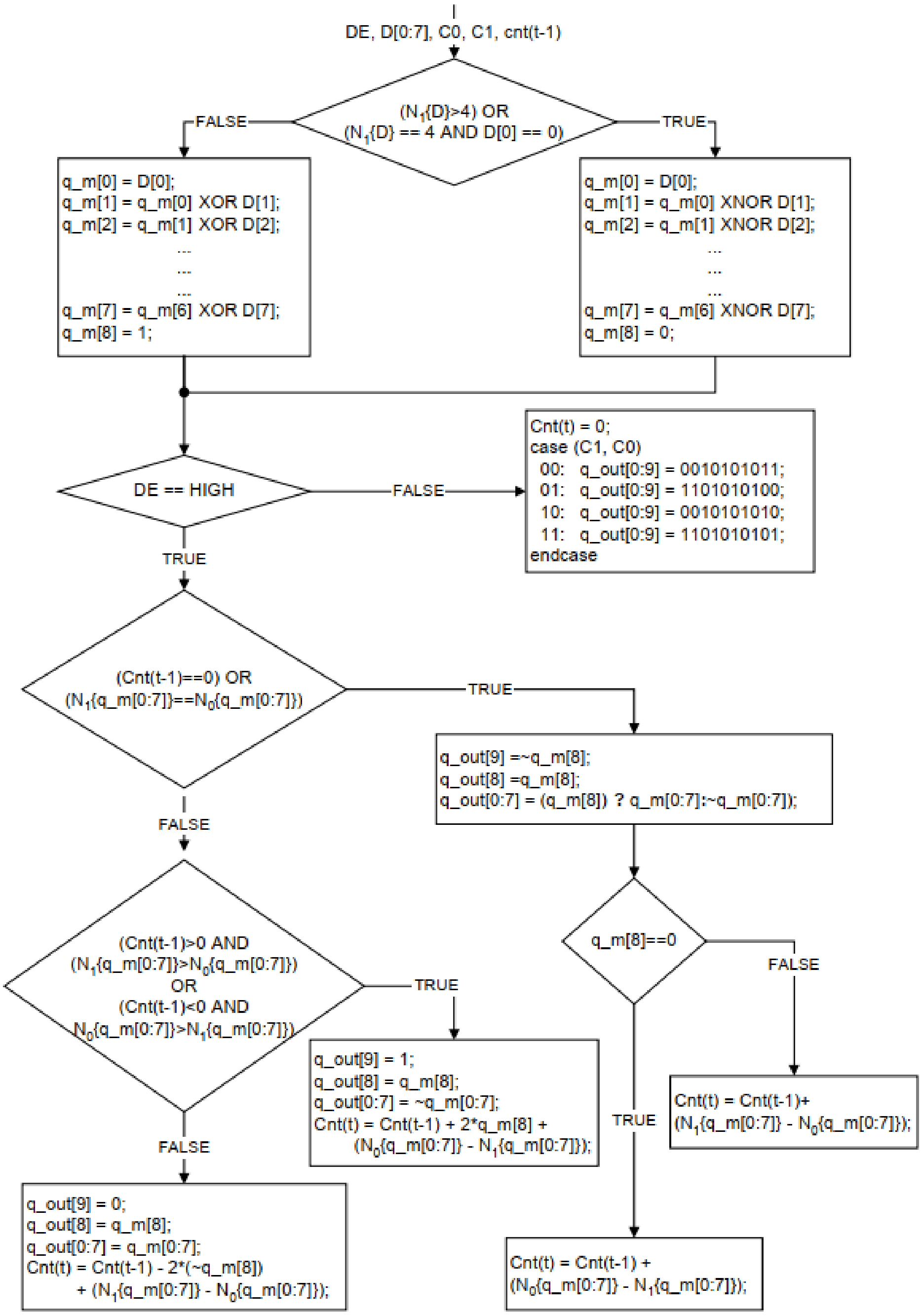


Table 3-1 Encoding Algorithm Definition

Signal	Description
D,C0,C1,DE	Input data for the encoding module. D represents 8-bit pixel data, C0 and C1 are control data for the corresponding channel, and DE is the data enable signal.
cnt	The register is used to track the disparity of data stream. A positive value represents the excess number of transmitted '1'. A negative value represents the excess number of transmitted "0". cnt {t-1} represents the previous disparity value of the previous input data. cnt {t} represents the new disparity value for the current input data.
q_out	Q_out is a 10-bit encoded output value.
N_1\{x\}	Returns the number of "1" in the array x.
N_0\{x\}	Returns the number of "0" in the array x.

After encoding, the 8-bit video data is converted into 10-bit data, and then the parallel data is converted into serial data by the serializer OSER10. Bit 0 is transmitted first.

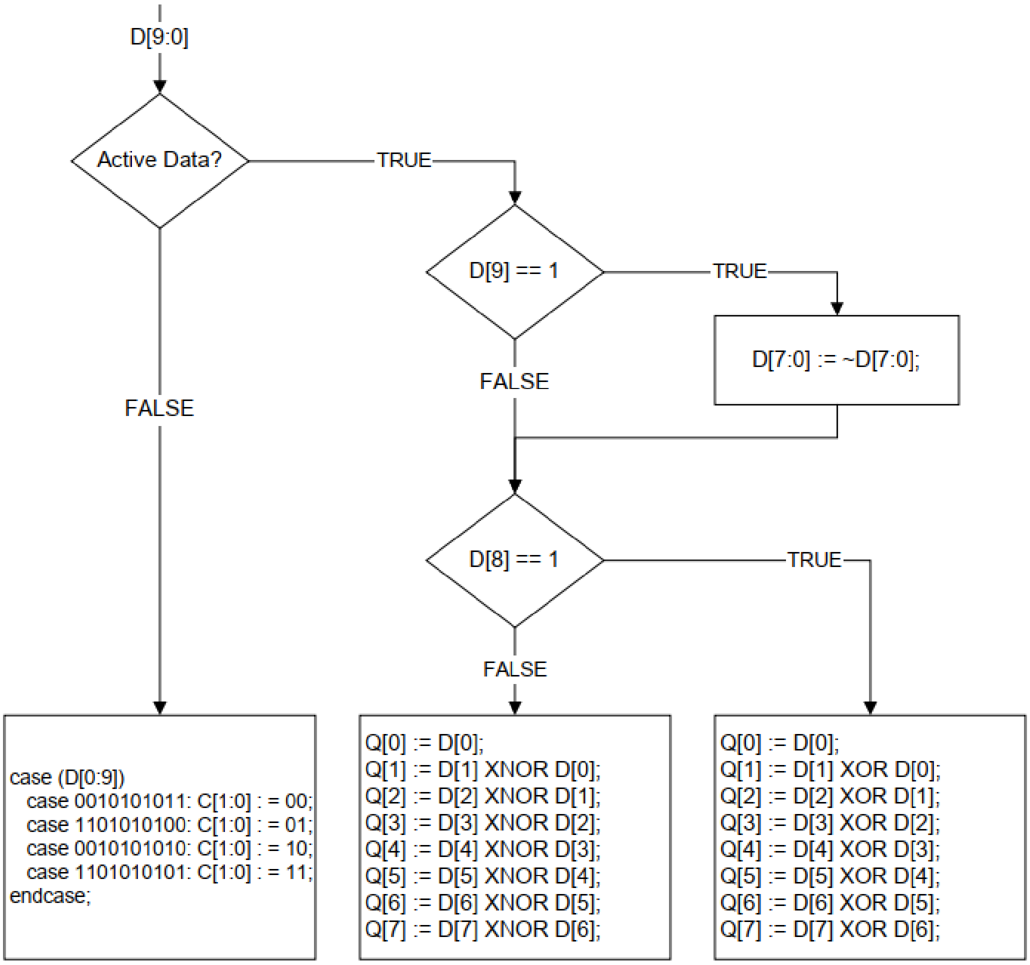
3.2.2 DVI RX

DVI RX first recovers the pixel clock from the clock channel and generates a five times serial clock. Then the serial data is converted into 10-bit parallel data via the deserializer IDES10.

Data synchronization alignment is performed based on the encoded values of HS and VS during the video signal blanking period, with the requirement that the length of each blanking period is at least greater than 128 character periods.

After data alignment, decoding is performed and the decoding algorithm is shown in Figure 3-4. The corresponding relationship between each channel and R, G and B is shown in Figure 3-2.

Figure 3-4 TMDS Decoding Algorithm Flow

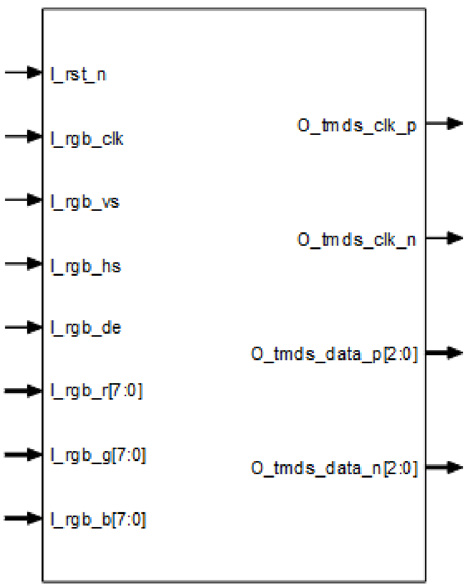


3.3 Port List

3.3.1 DVI TX

The I/O ports of Gowin DVI TX IP are shown in Figure 3-5.

Figure 3-5 DVI TX I/O Diagram



Ports vary slightly depending on the parameters.

The details of I/O ports of Gowin DVI TX IP are shown in Table 3-2.

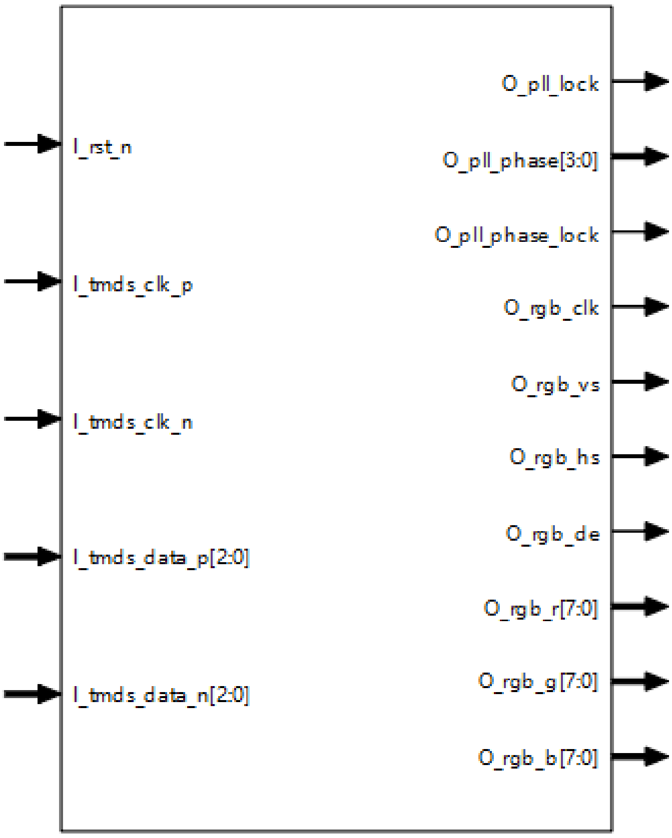
Table 3-2 I/O List of Gowin DVI TX IP

No.	Signal Name	I/O	Description	Remarks
1	I_rst_n	I	Reset signal, active low.	The I/O of all the signals takes DVI TX IP as reference.
2	I_serial_clk	I	This signal is valid when external clock is used I_serial_clk = I_rgb_clk * 5	
3	I_rgb_clk	I	Video input pixel clock	
4	I_rgb_vs	I	Video input field sync vs signal	
5	I_rgb_hs	I	Video input line sync hs signal	
6	I_rgb_de	I	Video input data enable de signal	
7	I_rgb_r	I	Video input data R component	
8	I_rgb_g	I	Video input data G component	
9	I_rgb_b	I	Video input data B component	
10	O_tmds_clk_p	O	Output TMDS differential signal clock positive	
11	O_tmds_clk_n	O	Output TMDS differential signal clock negative	
12	O_tmds_data_p	O	Output TMDS differential signal data positive ● Channel 0 corresponds to blue component ● Channel 1 corresponds to green component ● Channel 2 corresponds to red component	
13	O_tmds_data_n	O	Output TMDS differential signal data negative ● Channel 0 corresponds to blue component ● Channel 1 corresponds to green component ● Channel 2 corresponds to red component	

3.3.2 DVI RX

The I/O ports of Gowin DVI RX IP are shown in Figure 3-6.

Figure 3-6 DVI RX I/O Diagram



Ports vary slightly depending on the parameters.

The details of I/O ports of Gowin DVI RX IP are shown in Table 3-3.

Table 3-3 I/O List of Gowin DVI RX IP

No.	Name	I/O	Description	Remarks
1	I_rst_n	I	Reset signal, active low.	The I/O of all the signals takes DVI RX IP as reference.
2	I_tmds_clk_p	I	Input TMDS differential signal clock positive	
3	I_tmds_clk_n	I	Input TMDS differential signal clock negative	
4	I_tmds_data_p	I	Input TMDS differential signal data positive ● Channel 0 corresponds to blue component ● Channel 1 corresponds to green component ● Channel 2 corresponds to red component	
5	I_tmds_data_n	I	Input TMDS differential signal data negative ● Channel 0 corresponds to blue	

No.	Name	I/O	Description	Remarks
			component ● Channel 1 corresponds to green component ● Channel 2 corresponds to red component	
6	O_tmds_clk	O	● This signal is valid when external clock is used ● TMDS clock differential to single ended signal	
7	I_serial_clk	I	This signal is valid when external clock is used. ● I_serial_clk = O_tmds_clk * 5. ● The clock phase needs to be adjusted as required, and the default value is 90 degrees.	
8	O_pll_lock	O	● This signal is valid when external clock is not used. ● Internal PLL lock signal	
9	O_pll_phase	O	● This signal is valid when external clock is not used. ● Internal PLL output clock phase ● Correspondence between value and phase: 0:0.0, 1:22.5, 2:45, 3:67.5, 4:90, 5:112.5, 6:135, 7:157.5, 8:180, 9:202.5, 10:225, 11:247.5, 12:270, 13:292.5, 14:315, 15:337.5	
10	O_pll_phase_loc k	O	● This signal is valid when external clock is not used. ● Internal PLL output clock phase lock	
11	O_datar_bf	O	Valid when Debug option is enabled, data before word alignment, red component.	
12	O_datag_bf	O	Valid when Debug option is enabled, data before word alignment, green component.	
13	O_datab_bf	O	Valid when Debug option is enabled, data before word alignment, blue component.	
14	O_rgb_clk	O	Video output pixel clock	
15	O_rgb_vs	O	Video output field sync vs signal	
16	O_rgb_hs	O	Video output line sync hs signal	
17	O_rgb_de	O	Video output data enable de signal	
18	O_rgb_r	O	Video output data R component	
19	O_rgb_g	O	Video output data G component	
20	O_rgb_b	O	Video output data B component	

3.4 Parameters Configuration

3.4.1 DVI TX Parameters

Table 3-4 DVI TX Parameters

No	Name	Range	Default	Description
1	Using External Clock	Yes/No	No	If this parameter is configured, the external serial clock I_serial_clk is used. Or it will be generated by PLL.
2	TX Clock In Frequency	10.0~80.0	40.000MHz	Input pixel clock frequency
3	IO Setting	TLVDS/ELVDS	TLVDS	The type of IO buffer

3.4.2 DVI RX Parameters

Table 3-5 DVI RX Parameters

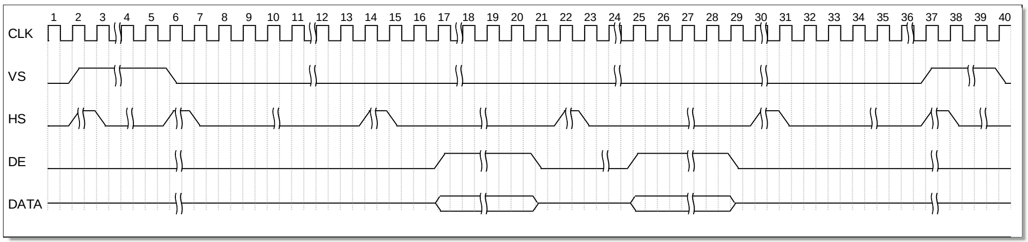
No.	Name	Range	Default	Description
1	Using External Clock	Yes/No	No	If this parameter is configured, the external serial clock I_serial_clk is used. Or it will be generated by PLL.
2	RX Clock In Frequency	10.0~80.0	40.000MHz	Input pixel clock frequency
3	Phase Search Mode	Auto/Manual	Auto	PLL output serial clock phase search mode ● Auto: Automatic mode ● Manual: Manual mode, and you need to input RX Clock Out Phase value
4	RX Clock Out Phase	0.0/22.5/45/67.5/90/112.5/135/157.5/180/202.5/225/247.5/270/292.5/315/337.5	90	Output serial clock phase value when Phase Search Mode is in manual mode
5	IO Delay Enable	Yes/No	No	IO Delay enable control
6	Channel0 IO Delay Value	0~127	0ps	IO delay control
7	Channel1 IO Delay Value	0~127	0ps	IO delay control
8	Channel2 IO Delay Value	0~127	0ps	IO delay control
9	Simulation Acceleration	Yes/No	No	● Debug option, simulation acceleration. ● If simulation is needed, enable this option. If simulation is not required, disable this option.
10	Data Before Align Enable	Yes/No	No	Debug option, data before word alignment enable.

3.5 Timing Description

This section describes the timing of Gowin DVI TX RX IP.

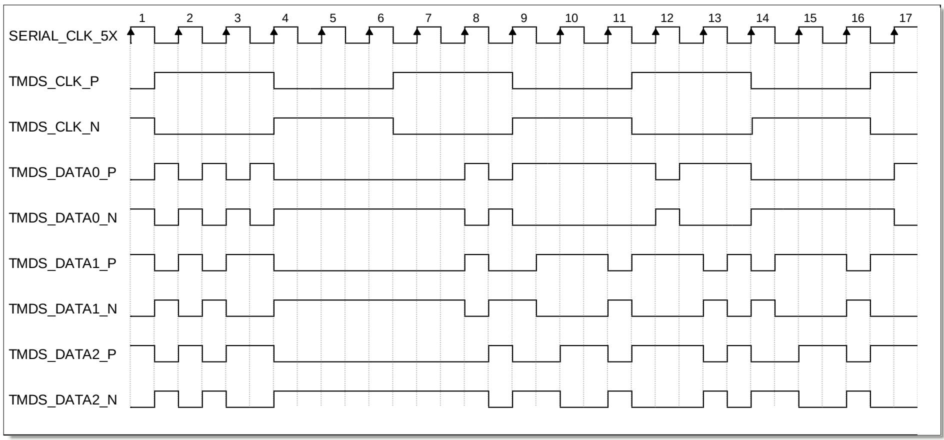
The timing diagram of DVI is as shown in Figure 3-7.

Figure 3-7 Timing Diagram of DVI



The timing diagram of TMDS is as shown in Figure 3-8.

Figure 3-8 Timing Diagram of TMDS



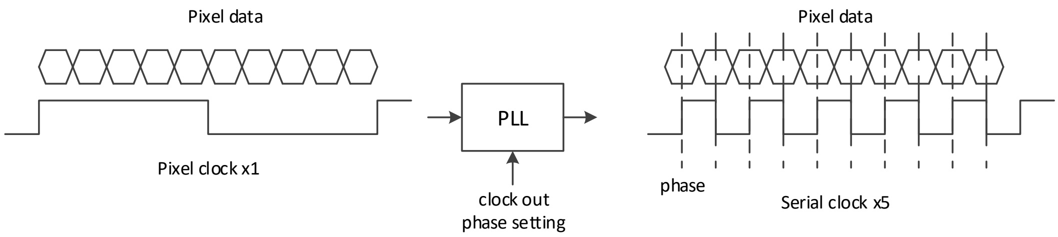
In DVI RX IP, you need to use TMDS_CLK, i.e. pixel clock, to generate five times serial clock for sampling serial data. Correct data sampling can only be achieved by setting an appropriate phase, as shown in Figure 3-9.

The Phase Search Mode option includes Auto and Manual.

- Auto: Users do not need to set it, it automatically searches for the proper phase.
- Manual: Users set the PLL phase by themselves on the GUI.

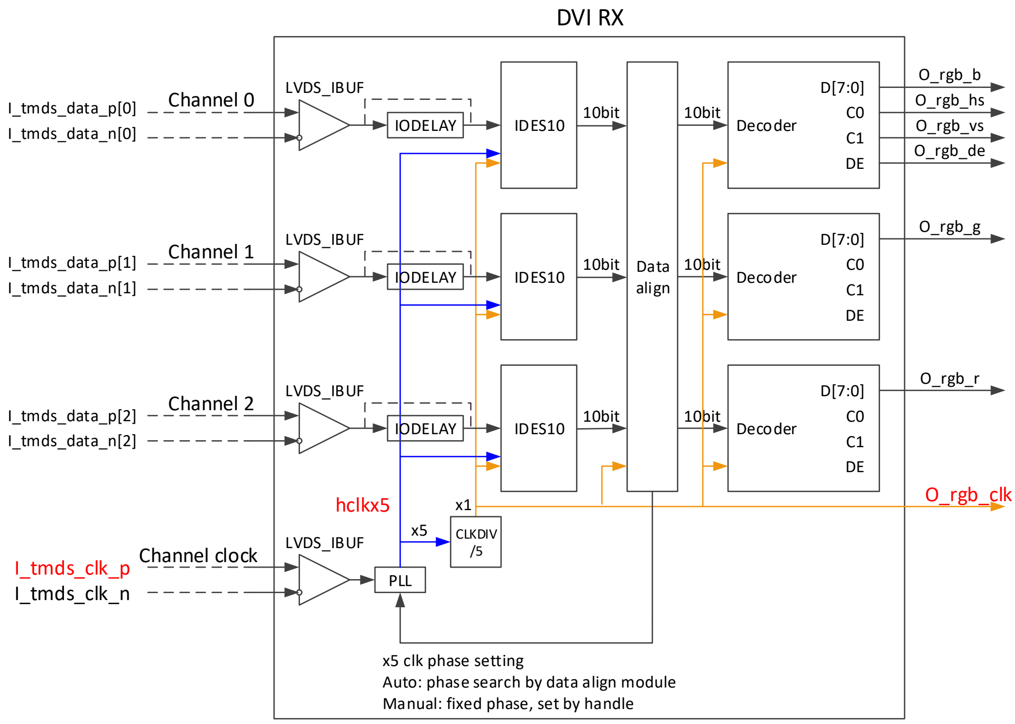
In manual mode, PLL lock is the fastest, it will lock when power on, provided that the user knows which phase is the best for data sampling. In Auto mode, PLL lock is slower, it may take a few seconds to search; if the cable is not good, or the signal quality is poor, it even will not search for the proper phase.

Figure 3-9 DVI RX Serial Data Sampling Diagram



In DVI RX IP, if the serial clock is generated using the internal PLL, the clock tree is as shown in Figure 3-10. The frequency of input differential clock I_tmds_clk is the frequency of pixel clock, and then it is converted from differential to single-ended and then multiplied by 5 using the PLL to obtain hclkx5. Finally, the output pixel clock O_rgb_clk is generated by dividing hclkx5 by 5 using CLKDIV.

Figure 3-10 Internal Clock Tree Diagram for DVI RX



It is recommended to apply timing constraints on the relevant clocks in the .sdc timing constraint file. For example, if the pixel clock frequency is 79.5 MHz, apply the following constraints to the relevant clocks (the clock signal name and module name depend on the actual names used in the project).

create_clock -name I_tmds_clk_p -period 12.579 -waveform {0 6.29} [get_ports {I_tmds_clk_p}] -add

create_clock -name O_rgb_clk -period 12.579 -waveform {0 6.29} [get_nets {O_rgb_clk}] –add

create_clock -name hclkx5 -period 2.516 -waveform {0 1.258} [get_nets {DVI_RX_Top_inst/dvi2rgb_inst/hclkx5}] -add

set_clock_groups -exclusive -group [get_clocks {O_rgb_clk}] -group [get_clocks {hclkx5}] -group [get_clocks {I_tmds_clk_p}]

4 Interface Configuration

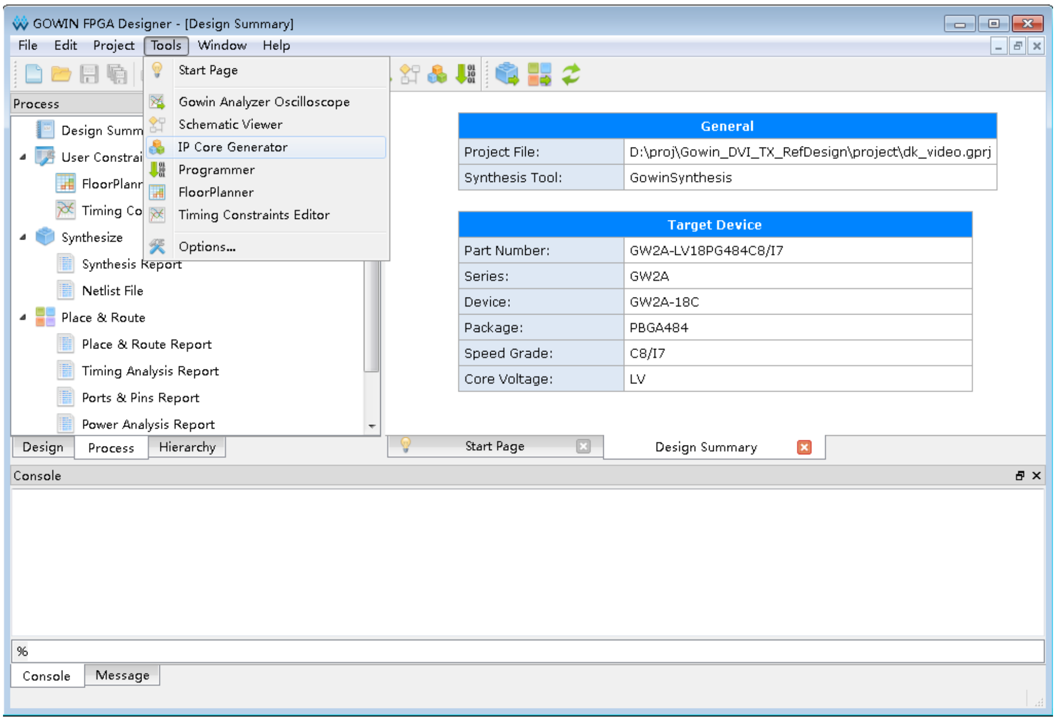
You can use IP core generator in the IDE to invoke and configure Gowin DVI TX RX IP.

4.1 DVI TX IP Configuration

1. Open IP Core Generator

After creating the project, click the "Tools" tab, select and open the "IP Core Generator" from the drop-down list, as shown in Figure 4-1.

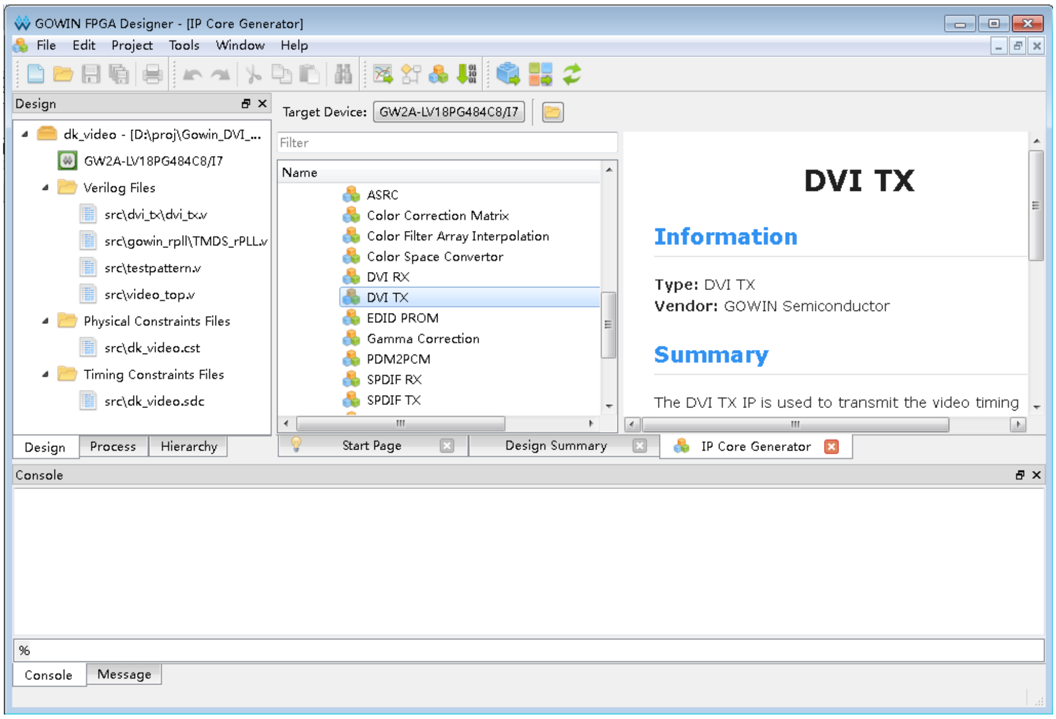
Figure 4-1 Open IP Core Generator



2. Open DVI TX IP Core

Click "Multimedia" and double click "DVI TX" to open the configuration interface, as shown in Figure 4-2.

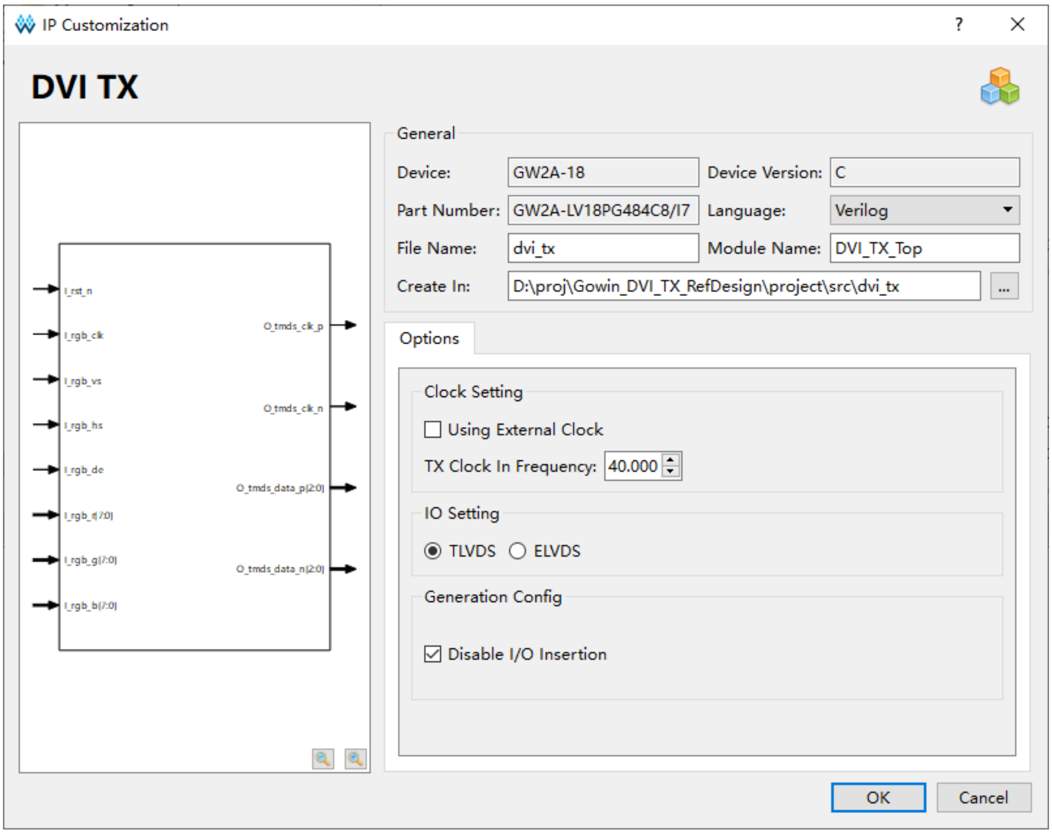
Figure 4-2 Open DVI TX IP Core



3. DVI TX IP Core Port Diagram

On the left of the configuration interface is the port diagram of DVI TX IP core, as shown in Figure 4-3.

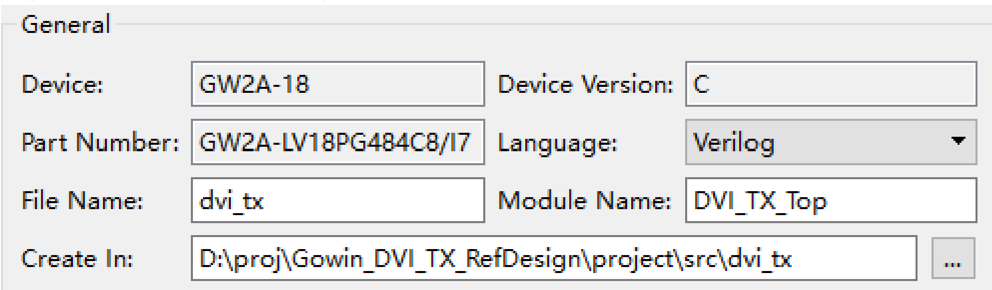
Figure 4-3 Port Diagram of DVI TX IP



4. General Configuration

See the project information in the upper part of the GUI. Take GW2A-18C chip as an example, and select PBGA484. The top-level file name of the generated project is shown in the "Module Name", and the default is "DVI_TX_Top", you can modify it as required. The folder generated by the IP core is shown in "File Name", which contains the files required by DVI TX IP core, and the default is "dvi_tx ", you can modify it as required. "Creat In" shows the path of generated folder. The default is "\project path\ src\dvi_tx", you can modify it as required.

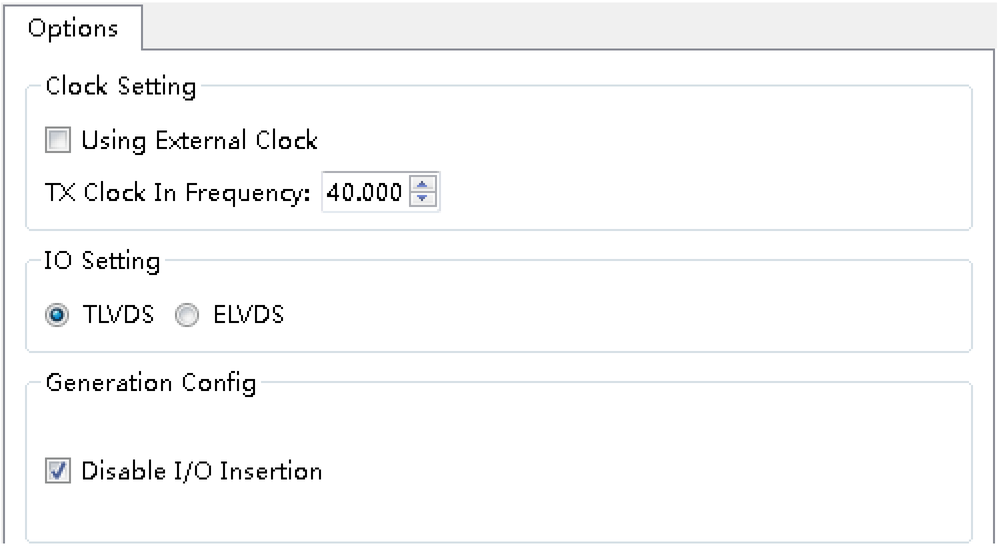
Figure 4-4 General Configuration



5. Options

You can configure clock parameters of DVI TX in "Clock Setting" tab.

Figure 4-5 Options

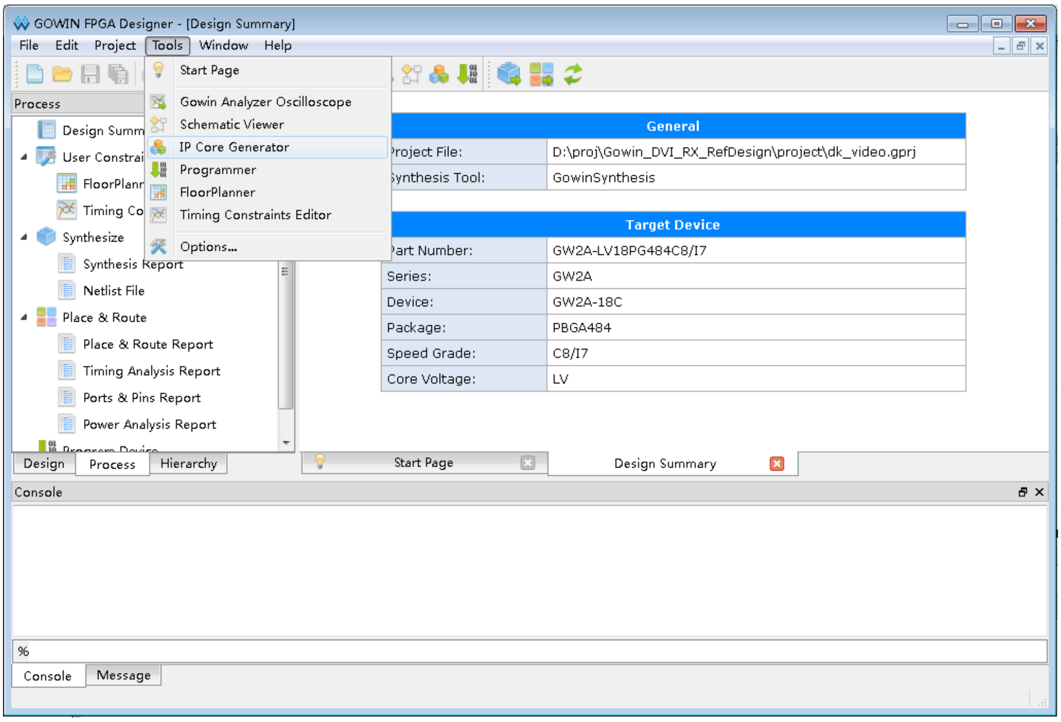


4.2 DVI RX IP Configuration

1. Open IP Core Generator

After creating the project, click the "Tools" tab, select and open the "IP Core Generator" from the drop-down list, as shown in Figure 4-6.

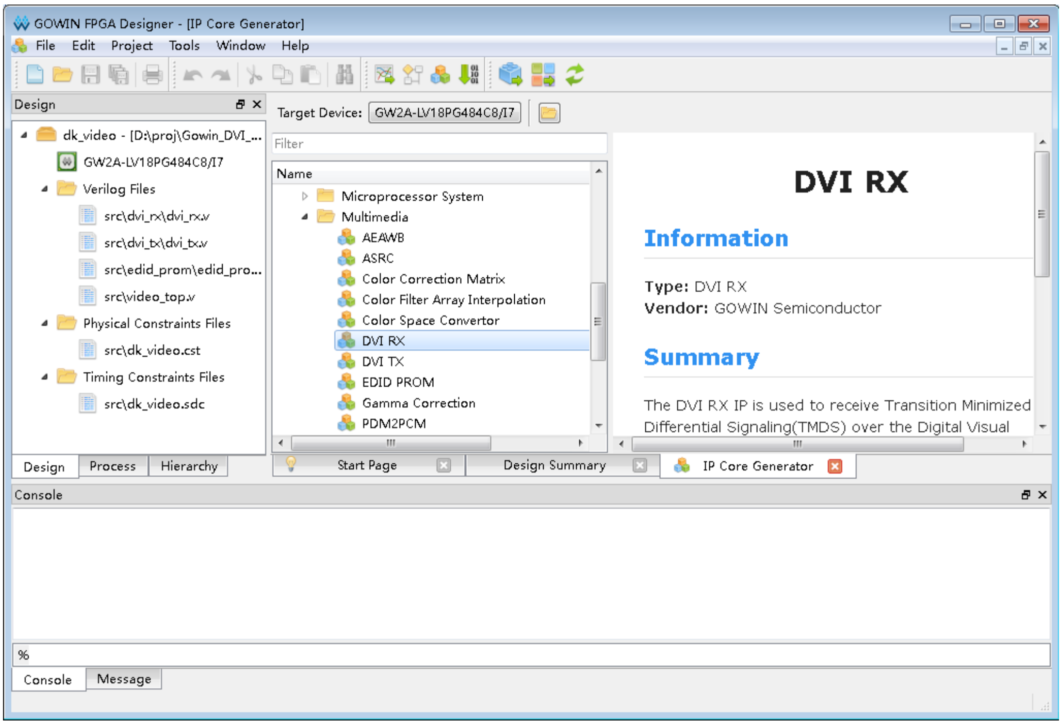
Figure 4-6 Open IP Core Generator



2. Open DVI RX IP Core

Click "Multimedia" and double click "DVI RX" to open the configuration interface, as shown in Figure 4-7.

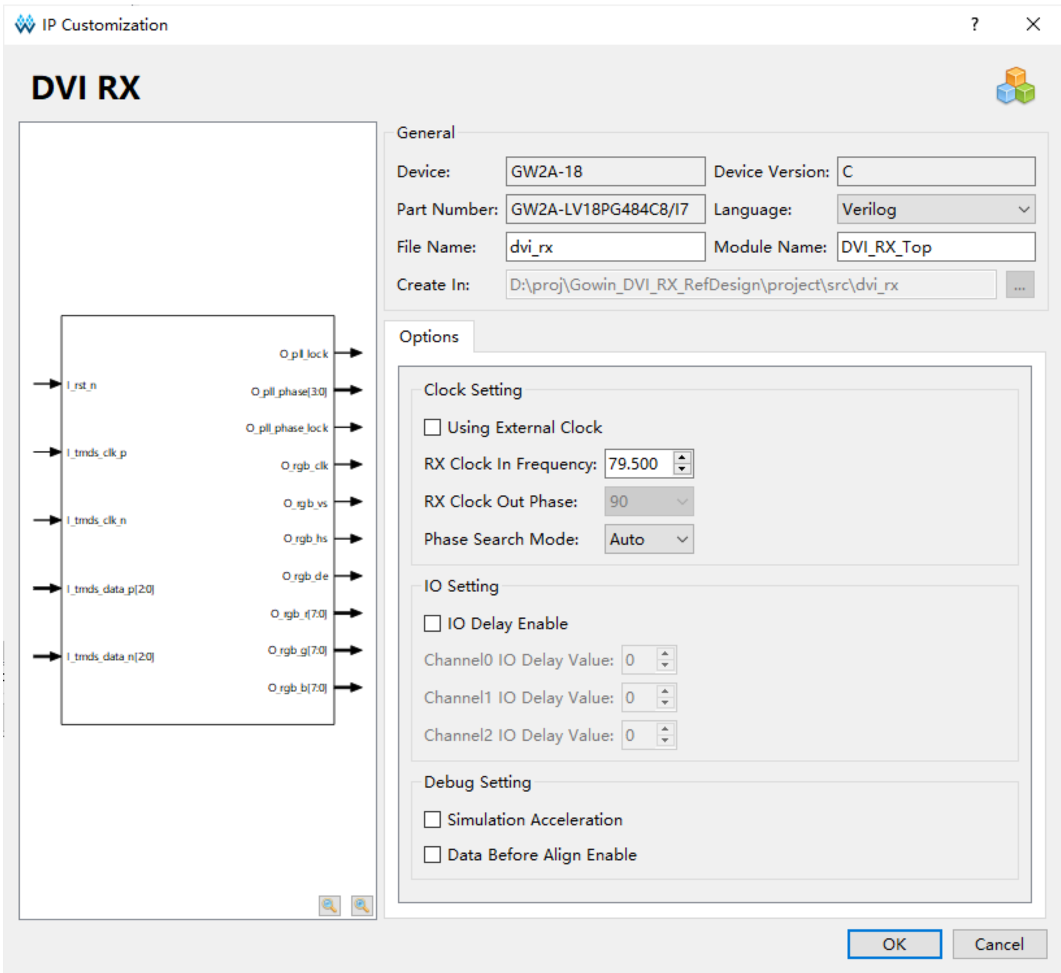
Figure 4-7 Open DVI RX IP Core



3. DVI RX IP Core Port Diagram

On the left of the configuration interface is the port diagram of DVI RX IP core, as shown in Figure 4-8.

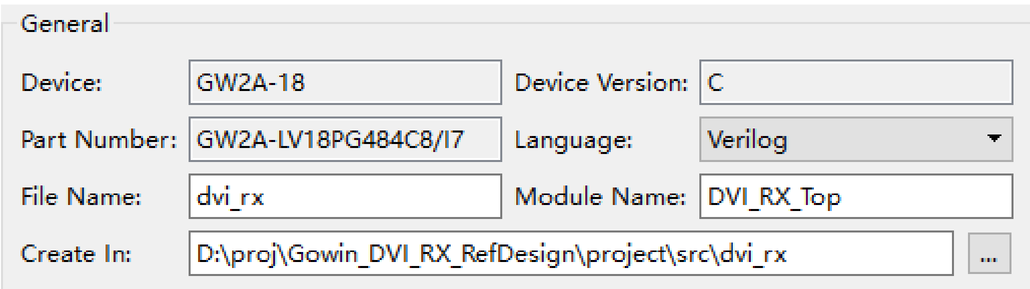
Figure 4-8 Port Diagram of DVI RX IP



4. General Configuration

See the project information in the upper part of the GUI. Take GW2A-18C chip as an example, and select PBGA484. The top-level file name of the generated project is shown in the "Module Name", and the default is "DVI_RX_Top", you can modify it as required. The folder generated by the IP core is shown in "File Name", which contains the files required by DVI RX IP core, and the default is "dvi_rx ", and you can modify it as required. "Creat In" shows the path of the generated folder. The default is "\project path\src\dvi_rx", and you can modify it as required.

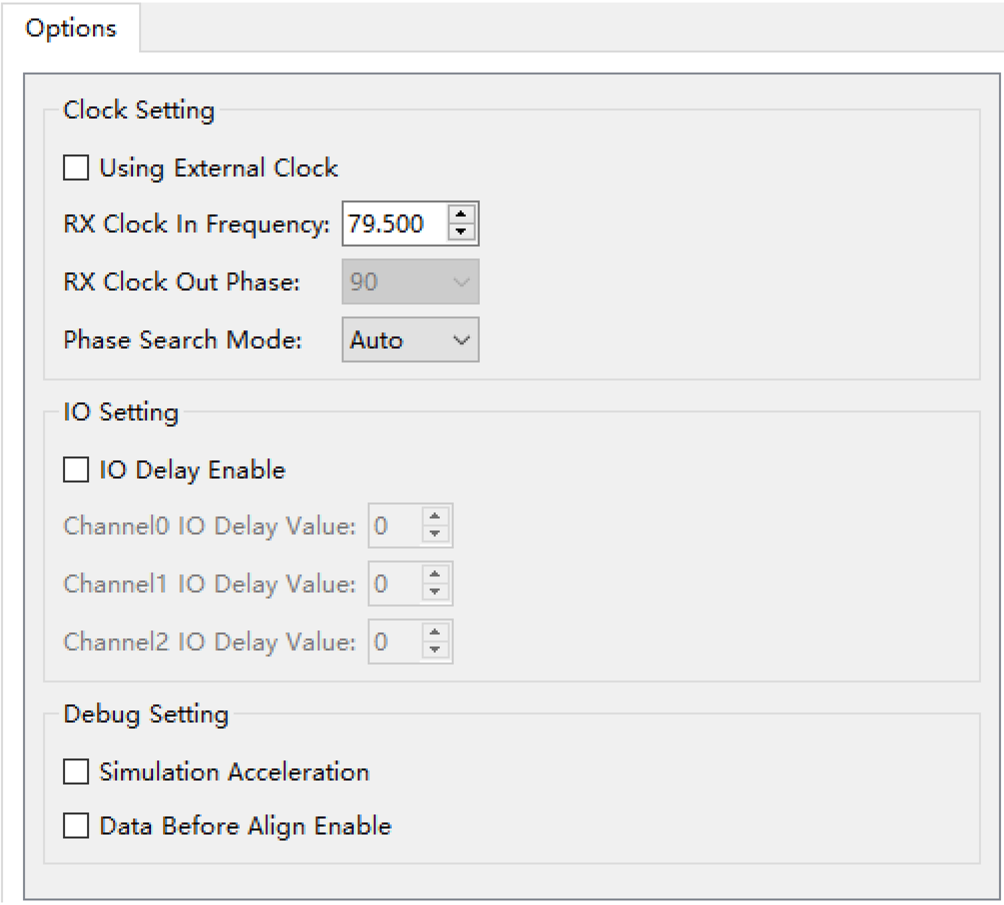
Figure 4-9 General Configuration



5. Options

You can configure clock parameters of DVI RX in "Clock Setting" tab and can configure channel IO delay value in "Data Setting" tab.

Figure 4-10 Options



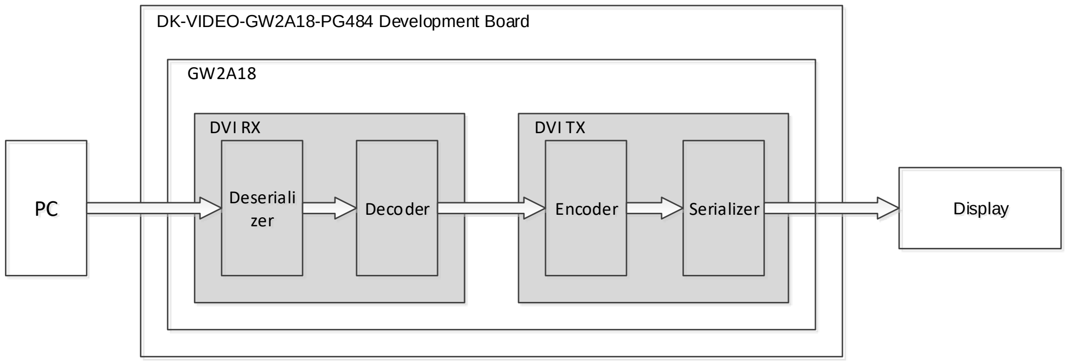
5 Reference Design

This chapter describes the usage of the reference design of Gowin DVI TX RX IP. See the reference design for details at Gowinsemi website.

5.1 Design Instance 1

Take DK-VIDEO-GW2A18-PG484 as an example, you can click here for the details of the board, and the diagram is as shown in Figure 5-1.

Figure 5-1 Diagram of Reference Design 1



In this instance, there are DVI RX IP and DVI TX IP, and the steps are as follows:

1. The project has included the EDID_PROM module and the 128-byte EDID file, and the recommended resolution is 1280x768.
2. The display device is connected to the PC via the HDMI3 RX interface with an HDMI cable, and the DVI format video with a resolution of 1280x768 will be output after the PC detects the connection of the display and recognizes the EDID information.
3. The TMDS signal of DVI is decoded by the DVI RX IP module. The decoded data is parallel video data.
4. The DVI TX IP module is used to encode the parallel data into TMDS signals.
5. Then output via HDMI4 TX interface. The monitor can display the output from the computer using HDMI cable.

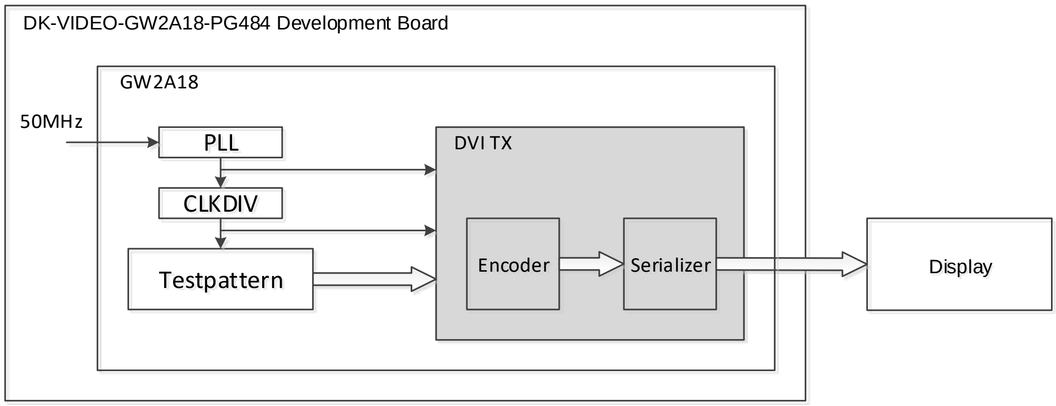
When the reference design is applied to board-level test, you can output the encoded signal to the display, or observe the data with the on-line logic analyzer or oscilloscope.

In the simulation project provided by the reference design, BMP bitmap is used as the test stimulus source, and tb is the top-level module of the simulation project. It can be compared with the output images after simulation.

5.2 Design Instance 2

Take DK-VIDEO-GW2A18-PG484 as an example, and the diagram is as shown in Figure 5-2.

Figure 5-2 Diagram of Reference Design 2



In this reference design, there is only DVI TX IP, and its steps are as follows:

1. The pixel clock and serial clock required by DVI TX IP is generated from 50 MHz reference clock.
2. The DVI format video with 1280x 720 resolution is output from the Testpattern module.
3. The DVI TX IP module is used to encode the parallel video data into TMDS signals.
4. Then output via HDMI4 TX interface. The monitor can display the output from computer using HDMI cable.

When the reference design is applied to board-level test, users can output the encoded signal to the display, or observe the data with the on-line logic analyzer or oscilloscope.

In the simulation project provided by the reference design, BMP bitmap is used as the test stimulus source, and tb is the top-level module of the simulation project. It can be compared with the output images after simulation.

6 File Delivery

The delivery file of Gowin DVI TX RX IP includes documentation, source code and reference design.

6.1 Documents

The folder mainly contains the user guide in PDF.

Table 6-1 Documents List

Name	Description
IPUG938, Gowin DVI TX RX IP User Guide	Gowin DVI TX RX IP User Guide, namely this one.

6.2 Design Source Code (Encryption)

The encrypted code folder contains RTL encrypted code, which is used by the GUI to generate the IP core required by users.

Table 6-2 DVI TX Design Source Code List

Name	Description
dvi_tx.v	The top-level file of the IP core, which provides you with interface information, encrypted.

Table 6-3 DVI RX Design Source Code List

Name	Description
dvi_rx.v	The top-level file of the IP core, which provides users with interface information, encrypted.

6.3 Reference Design

Gowin DVI RX RefDesign folder contains the netlist file, reference design, constraints file, top-level file and the project file, etc.

Table 6-4 Gowin DVI RX RefDesign File List

Name	Description
video_top.v	The top module of reference design
dk_video.cst	Project physical constraints file
dk_video.sdc	Project timing constraints file
dvi_tx	DVI TX IP project folder
dvi_tx.v	Generates DVI TX IP top-level file, encrypted.

Name	Description
dvi_tx.vo	Generates DVI TX IP netlist file
dvi_rx	DVI RX IP project folder
dvi_rx.v	Generates DVI RX IP top-level file, encrypted.
dvi_rx.vo	Generates DVI RX IP netlist file

Gowin DVI TX RefDesign folder contains the netlist file, reference design, constraints files, top-level files and the project folder, etc.

Table 6-5 Gowin DVI TX RefDesign File List

Name	Description
video_top.v	The top module of reference design
dk_video.cst	Project physical constraints file
dk_video.sdc	Project timing constraints file
testpattern.v	Reference design file
dvi_tx	DVI TX IP project folder
dvi_tx.v	Generates DVI TX IP top-level file, encrypted.
dvi_tx.vo	Generates DVI TX IP netlist file
gowin_rpll	PLL project folder
TMDS_rPLL.v	Generates phase-locked loop top-level file

7 Appendix

7.1 DVI and HDMI Compatibility

HDMI is developed on the basis of DVI with TMDS, which is an extension of DVI. Therefore, except for audio signals and related control signals, DVI and HDMI are compatible on the interface.

7.2 DVI Pinout

The DVI-D connector diagram is as shown in Figure 7-1 and the pinout is as shown in Table 7-1. When the DVI signal is single link, TMDS data 3±, TMDS data 4± and TMDS data 5± are not used.

Figure 7-1 DVI-D Connector

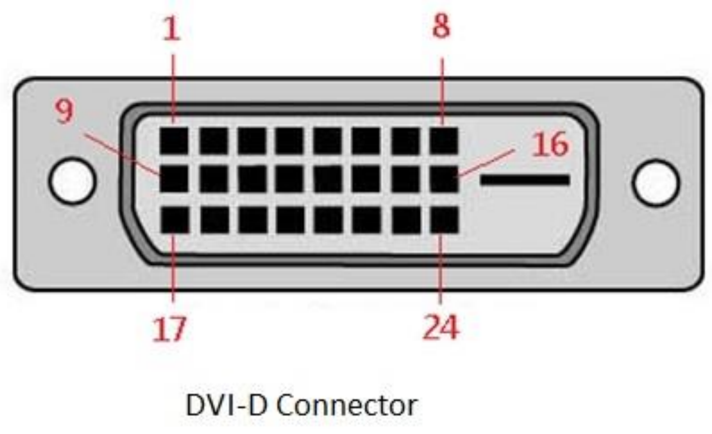


Table 7-1 Pinout of DVI-D Connector

Pin	Description
1	TMDS data 2-
2	TMDS data 2+
3	TMDS data 2/4 shield
4	TMDS data 4-
5	TMDS data 4+
6	DDC clock
7	DDC data
8	No connection

Pin	Description
9	TMDS data 1-
10	TMDS data 1+
11	TMDS data 1/3 shield
12	TMDS data 3-
13	TMDS data 3+
14	+5V DC power
15	Ground（+5V circuit）
16	Hot plug detection
17	TMDS data 0-
18	TMDS data 0+
19	TMDS data 0/5 shield
20	TMDS data 5-
21	TMDS data 5+
22	TMDS clock shield
23	TMDS clock+
24	TMDS clock-

7.3 HDMI Pinout

Type A HDMI connector diagram is as shown in Figure 7-2 and the pinout is as shown in Table 7-2.

Figure 7-2 Type A HDMI Connector

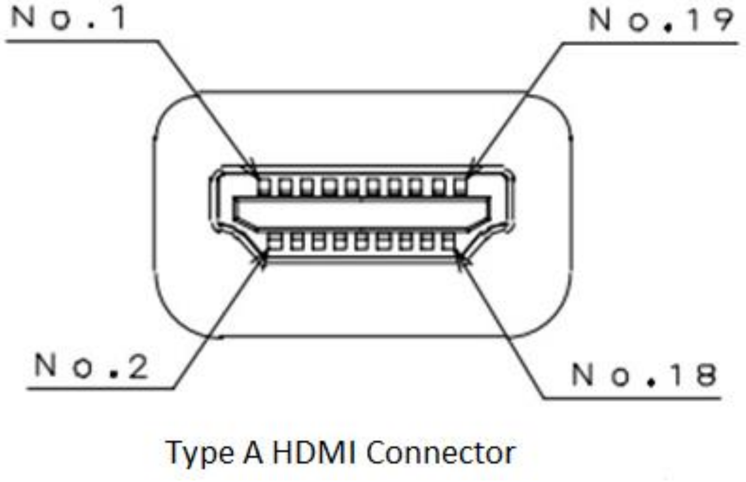


Table 7-2 Pinout of Type A HDMI Connector

Pin	Description
1	TMDS data 2+
2	TMDS data 2 shield
3	TMDS data 2-
4	TMDS data 1+
5	TMDS data 1 shield
6	TMDS data 1-

Pin	Description
7	TMDS data 0+
8	TMDS data 0 shield
9	TMDS data 0-
10	TMDS clock+
11	TMDS clock shield
12	TMDS clock-
13	CEC
14	Reserved
15	SCL
16	SDA
17	DDC/CEC Ground
18	+5V DC power
19	Hot plug detection

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